

PCA

x	y	$(x - \bar{x})$	$(y - \bar{y})$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
2.5	2.4	0.69	0.49	0.3381	0.4761	0.2401
0.5	0.7	-1.31	-1.21	1.5851	1.7161	1.4641
2.2	2.9	0.39	0.99	0.3861	0.1521	0.9801
1.9	2.2	0.09	0.29	0.0261	0.0081	0.0841
3.1	3.0	1.29	1.09	1.4061	1.6641	1.1881
2.3	2.7	0.49	0.79	0.3871	0.2401	0.6241
2	1.6	0.19	-0.31	-0.0589	0.0361	0.0961
1	1.1	-0.81	-0.81	0.6561	0.6561	0.6561
1.5	1.6	-0.31	-0.31	0.0961	0.0961	0.0961
1.1	0.9	-0.71	-1.01	0.7171	0.5041	1.0201
<u>$\bar{x} = 1.81$</u> <u>$\bar{y} = 1.91$</u>				<u>5.539</u>	<u>5.549</u>	<u>6.449</u>

Covariance Matrix

$$C = \begin{bmatrix} \overset{0.617}{\text{cov}(x, x)} & \overset{0.6154}{\text{cov}(x, y)} \\ \underset{0.6154}{\text{cov}(y, x)} & \underset{0.716}{\text{cov}(y, y)} \end{bmatrix}$$

$$\frac{5.549}{9} \quad \text{cov}(x, x) = \frac{\sum (x - \bar{x})(x - \bar{x})}{n-1} \quad \boxed{0.617}$$

$$\frac{5.539}{9} \quad \text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1} \quad \boxed{0.6154}$$

$$\frac{6.449}{9} \quad \text{cov}(y, y) = \frac{\sum (y - \bar{y})(y - \bar{y})}{n-1} \quad \boxed{0.716}$$

Step 2 = Calculate Eigen values.

$$|C - \lambda I| = 0$$

$$\left| \begin{bmatrix} 0.617 & 0.6154 \\ 0.6154 & 0.716 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 0.617 & 0.6154 \\ 0.6154 & 0.716 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$\left| \begin{bmatrix} 0.617 - \lambda & 0.6154 \\ 0.6154 & 0.716 - \lambda \end{bmatrix} \right|$$

$$(0.617 - \lambda)(0.716 - \lambda) - (0.6154)^2$$

$$\lambda^2 - (0.617\lambda) - 0.716\lambda - (0.378)$$

$$\boxed{\lambda^2 - 1.33\lambda + 0.063 = 0} \quad +0.4417$$

$$\underline{\lambda_1 = 1.2808}$$

$$\underline{\lambda_2 = 0.0491}$$

Step 3 = Eigen vectors

$$CV_1 = \underline{N} V_1$$

$$\begin{bmatrix} \underline{0.617} & \underline{0.6154} \\ \underline{0.6154} & \underline{0.716} \end{bmatrix} * \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \underline{1.2808} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$0.617 x_1 + 0.6154 y_1 = 1.2808 x_1$$

$$0.617 x_1 - 1.2808 x_1 = -0.6154 y_1$$

$$\boxed{-0.6638 x_1 = -0.6154 y_1}$$

$$0.6154 x_1 + 0.716 y_1 = 1.2808 y_1$$

$$\begin{aligned} 0.6154 x_1 &= 1.2808 y_1 - 0.716 y_1 \\ &= \boxed{0.6154 x_1 = 0.6654 y_1} \end{aligned}$$

$$-0.6638x_1 = 0.615y_1$$

$$x_1 = 0.9270$$

$$v_1 = \begin{bmatrix} 0.927 \\ 1 \end{bmatrix} \quad \frac{0.927}{\sqrt{(0.927)^2 + (1)^2}}$$

$$v_1^* = \begin{bmatrix} 0.679 \\ 0.733 \end{bmatrix} = \frac{0.679}{\sqrt{(0.927)^2 + (1)^2}} \\ = 0.733$$

$$Cv_2 = N_2 v_2$$

$$\begin{bmatrix} 0.617 & 0.6154 \\ 0.6154 & 0.716 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = 0.0491 \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$0.617x_2 + 0.6154y_2 = 0.0491x_2$$

$$\begin{cases} 0.6154x_2 + 0.716y_2 = 0.0491y_2 \end{cases}$$

$$\rightarrow 0.617x_2 - 0.0491x_2 = -0.6154y_2$$

$$\boxed{0.5679x_2 = -0.6154y_2}$$

$$0.6154x_2 = 0.0491y_2 - 0.716y_2$$

$$0.6154x_2 = -0.6669y_2$$

$$0.5679x_2 = -0.6154y_2$$

$$x_2 = -0.6154/0.5679$$

$$x_2 = -1.0836$$

$$v_2 \begin{bmatrix} -1.0836 \\ 1 \end{bmatrix}$$

$$\frac{-1.0836}{\sqrt{-1.0836^2 + 1^2}}$$

$$v_2^* \begin{bmatrix} -0.7348 \\ 0.678 \end{bmatrix} = \frac{1}{\sqrt{-1.0836^2 + 1^2}}$$

x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

$$\begin{bmatrix} 0.679 \\ 0.733 \end{bmatrix}$$

Rf	λ_1
3.45	$\boxed{1.2808}$
0.85	λ_2
3.475	λ_2
2.9	λ_1
4.40	λ_1
3.53	λ_2
2.52	λ_2
1.48	λ_1
2.18	λ_2
1.40	λ_1

$$\lambda_1 \begin{bmatrix} 0.679 \\ 0.733 \end{bmatrix}$$

$$\lambda_2 \begin{bmatrix} -0.7348 \\ 0.678 \end{bmatrix}$$

Moving Average.

		<u>MA3</u>	M.A5	7	5.5/3
2.5	—	—	—	—	—
0.5	—	—	—	—	—
2.2	1.833	—	—	—	4.15/3
1.9	1.53	—	—	—	7.2/3
3.1	2.4	2.04	—	—	7.3/3
2.3	2.43	—	—	—	10.2/5
2	—	—	—	—	—
1	—	—	—	—	—
1.5	—	—	—	—	—
1.1	—	—	—	—	—